

ECE 3340 Numerical Methods - Exam 1A

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Binary Numbers

_____ / 40

signed integers _____ / 10

unsigned integers _____ / 5

floating point _____ / 20

Catastrophic Cancellation

_____ / 30

Mean Value Theorem _____ / 10

Precision and Error _____ / 20

Solving Equations

_____ / 30

Newton's Method _____ / 20

False Position _____ / 10

Total

_____ / 100

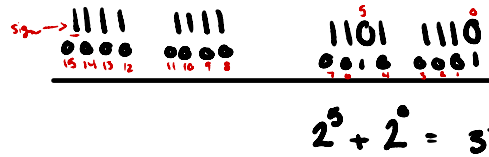
notes on:
C.C
MVT

FLIP THE BITCH

[10 points] Convert the following 16-bit signed integer register to decimal:

DE FF
FF DE

- 33



[5 points] What are the minimum and maximum values of a signed 6-bit integer?

6 bit signed range = -2^{n-1} to 2^{n-1} with $n=6$

$$-2^5 = -32$$

$$2^5 - 1 = 31$$

min:

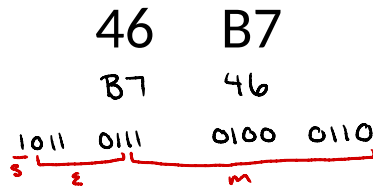
-32

max:

31

[20 points] Provide the decimal value stored in these 16-bit floating point registers assuming a 5-bit exponent and a bias of 15:

This means that the hex given is what is stored, so we need to flip it to achieve the dec. value.
little endian form



-0.4545

Exponent: $01101 \rightarrow 13 - 15 = -2$

$$1.1101000110 \rightarrow -1.818_{10} \times 2^{-5} \rightarrow -0.4545$$

[10 points] Use the Mean Value Theorem to improve precision in the following expression. Put the reformulation in the box provided.

MVT $\rightarrow f(b) - f(a) = f'(c)(b-a)$

$$\cos 1.11 - \cos 1.1$$

$$\rightarrow 1.1 - 1.11 = -0.01$$

$$c = \frac{1.1 + 1.1}{2} = 1.105$$

$$(b-a)(-sin c) \rightarrow (1.11 - 1.1)(-sin(1.105))$$

$$\text{reformulated: } \cos 1.11 - \cos 1.1 \approx -0.01 \sin(1.105)$$

-0.01 sin(1.105)

[20 points] The following function $f(x)$ will exhibit catastrophic cancellation as $x \rightarrow 0$. Calculate an alternative function $g(x)$ that improves precision (using any combination of reformulation, Taylor series, or the mean

value theorem). Demonstrate the improvement by carrying out the calculations for $f(0.5)$ using **3 digits of precision**. Provide the corresponding relative errors.

$$f(x) = \cos(x) - \sqrt{1-x^2}$$

Method 1

Taylor Series

① Taylor of $\cos(x)$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \frac{x^8}{40320} - \dots$$

② Taylor of $\sqrt{1-x^2}$

$$\sqrt{1-x^2} = 1 - \frac{x^2}{2} - \frac{x^4}{8} - \frac{x^6}{16} - \frac{5x^8}{128} - \dots$$

③

$$\left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \frac{x^8}{40320}\right) - \left(1 - \frac{x^2}{2} - \frac{x^4}{8} - \frac{x^6}{16} - \frac{5x^8}{128}\right)$$

④ True value **BIG NOTE PUT CAL IN RAD**

$$f_{true}(0.5) = \cos(0.5) - \sqrt{1-(0.5)^2} =$$

$$0.877 - 0.866 \rightarrow 0.11557 \leftarrow f_{true} = 0.11557$$

Before + After

⑤

$$g(x) = \frac{x^4}{6} + \frac{11x^6}{180} + \frac{197x^8}{5040}$$

$$g(0.5) = \frac{(0.5)^4}{6} + \frac{11(0.5)^6}{180} + \frac{197(0.5)^8}{5040} = 0.11524$$

$x=0.5$

$f(0.5) = 0.0115$

THIS WILL BE PUT INTO 3 SIG FIGS

⑥

$$\cos(0.5) = 0.878$$

$$\sqrt{1-(0.5)^2} = 0.866$$

$$0.878 - 0.866 = 0.012 \rightarrow f(0.5) = 0.012$$

$E_r = 0.49\%$

⑦ Error of Before and After

B:

$$\frac{|0.012 - 0.011557|}{|0.011557|} \times 100 \rightarrow 3.83\%$$

After

$$\frac{0.0115 - 0.011557}{0.011557} \times 100 = 0.49\%$$

Method 2

Conjugate

① Conjugate of the equation

$$\begin{aligned} \cos(x) - \sqrt{1-x^2} &= \left(\frac{\cos(x) + \sqrt{1-x^2}}{\cos(x) + \sqrt{1-x^2}} \right) \\ \cos^2 x &= 1 - \sin^2 x \quad \text{trig identities} \\ \sin^2 x &= 1 - \cos^2 x \\ &= \frac{\cos^2(x) - \sqrt{1-x^2}}{\cos(x) + \sqrt{1-x^2}} \rightarrow \frac{1 - \sin^2(x) - (1-x^2)}{\cos(x) + \sqrt{1-x^2}} \\ &= \frac{x^2 - \sin^2 x}{\cos(x) + \sqrt{1-x^2}} \end{aligned}$$

② Expected Results:

$$\cos(0.5) - \sqrt{1-(0.5)^2} \rightarrow 0.0116$$

③ $f(0.5)$

$$\cos(0.5) - \sqrt{1-(0.5)^2} =$$

$$0.878 - 0.866 = 0.012$$

④ $f_{approx}(0.5)$

$$\frac{(0.5)^2 - \sin^2(0.5)}{\cos(0.5) + \sqrt{1-(0.5)^2}} = 0.011557$$

⑤ $Error_B \leftarrow Error_A$

$Error_B$

$$\frac{0.011557 - 0.012}{0.011557} \times 100 = 3.83\%$$

$Error_A$:

$$\frac{0.011557 - 0.0115}{0.011557} \times 100 = 0.495\%$$

Newton's general formula = $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

[20 points] Solve the following equation using 5 iterations of Newton's method with an initial guess of $x_0 = 0.5$ and 3 digits of precision:

Making an iteration table!!!!!!

$$e^x \cos x = \sin x$$

$$f(x) = e^x \cos x - \sin x$$

$$f'(x) = (e^x - 1) \cos(x) - e^x \sin(x)$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

	x_n	$h(x_n)$	$h'(x_n)$	x_{n+1}
0.	0.5	$e^{0.5} \cos(0.5) - \sin(0.5) = 0.967$	$(e^{0.5} - 1) \cos(0.5) - e^{0.5} \sin(0.5) = -0.221$	$0.5 - \frac{0.967}{-0.221} \rightarrow 4.88$
1.	4.88	$e^{4.88} \cos(4.88) - \sin(4.88) = 22.9$	$(e^{4.88} - 1) \cos(4.88) - e^{4.88} \sin(4.88) = 152$	$4.88 - \frac{22.9}{152} \rightarrow 4.73$
2.	4.73	$e^{4.73} \cos(4.73) - \sin(4.73) = 2.99$	$(e^{4.73} - 1) \cos(4.73) - e^{4.73} \sin(4.73) = 115$	$4.73 - \frac{2.99}{115} \rightarrow 4.704$
3.	4.704	0.0139	110	4.703
4.	4.703	-0.035	109	4.703

[10 points] The following is a plot of the polynomial

false position general formula

$$c = b - \frac{f(b)(b-a)}{f(b) - f(a)} = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

$$\frac{1}{20} (x+4)(x+2)(x+1)(x-1) + 2$$

on the interval $[-3, 3]$. Draw and label the first three steps of the method of false position to find the root at $x=1.5$ given the interval $[-3, 3]$.

I SHOW MY WORK

ON THE NEXT

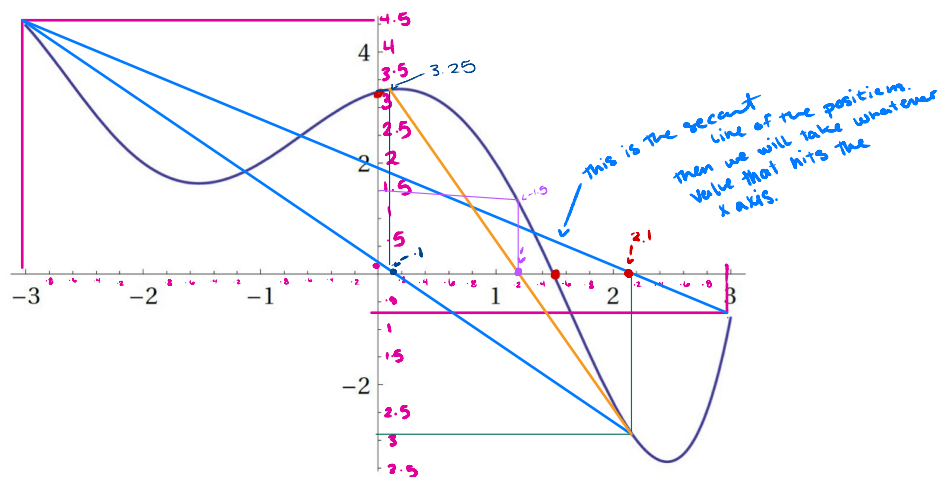
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false position general formula

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on the interval $[-3, 3]$. Draw and label the first three steps of the method of false position to find the root at $x=1.5$ given the interval $[-3, 3]$.



this may have approx. error but the method is the same as Roman's.

if $f(a) \cdot f(c) < 0$, $b_{n+1} = c_n$
 else if $f(a) \cdot f(c) > 0$, $a_{n+1} = c_n$

	a	f(a)	b	f(b)	c	f(c)
0	-3	4.5	3	0.75	2.1	-2.75?
1	-3	4.5	2.1	-2.75	0.1	3.25
2	0.1	3.25	2.1	-2.75	1.2	1.5
3						

← would the